

HAMZA MOORAJ

AI Researcher — Multimodal & Agentic Systems | Evaluation, Reliability, and Generalisation

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Applied AI researcher focused on evaluating and understanding the behaviour of multimodal and agentic systems under real-world constraints. Experience designing evaluation frameworks and analysing model reliability under distribution shift and complex task settings.

PUBLICATIONS & PROJECTS

A Systematic Exploration of Architectural Trade-offs for Crop Disease Classification - Lead Author

July 2025 – April 2026 — Under Review at **TLMR** (2026) — [Dataset released on HuggingFace](#) — [arXiv Preprint](#)

- Created AgriPath-LF16, a 65-class benchmark with explicit lab/field domain separation.
- Designed a unified evaluation framework comparing CNNs, contrastive VLMs, and generative VLMs.
- Analysed model behaviour under domain shift, revealing reliability gaps across architectures.
- Evaluated robustness differences between model families in controlled vs real-world settings.

Mechanistic Analysis of Induction in a Tiny Transformer - Personal Project

February 2026

- Trained a 2-layer transformer on a synthetic induction task to study learned behaviour.
- Analysed attention patterns to characterise offset-diagonal and recency-biased mechanisms.
- Designed interventions (ablation, rotation, perturbation) to probe causal structure.
- Identified shortcut learning behaviour in place of canonical induction mechanisms.

Medical Question-Answering (QA) Conversational Agent (Course: F21CA) - Project Lead

January 2025 – April 2025

- Developed a RAG-based medical QA system over 411 NHS records.
- Designed a tri-modal interaction pipeline integrating text, speech, and vision.
- Evaluated retrieval-augmented vs baseline performance using semantic similarity metrics.
- Demonstrated measurable reliability improvement from retrieval grounding (+5.6%).

EXPERIENCE

AI “Learning Buddy” Project Intern — Heriot-Watt University

July 2025 – September 2025

- Built a retrieval-based learning assistant using LlamaIndex and Milvus over course materials.
- Designed ingestion and chunking strategies to improve retrieval relevance across domains.
- Analysed embedding and retrieval behaviour to optimise response quality.
- Deployed and tested the system in a controlled university environment.

Data Science & Machine Learning Intern — Love For Data

August 2023

- Designed a popularity index across 1,970 Play Store apps spanning US and Pakistan datasets.
- Modelled feature–outcome relationships using a Random Forest regressor, analysing performance drivers.

Software Engineering Intern — Folio3

June 2023 – July 2023

- Contributed to a Visitor Management System within a sprint-based development team.

EDUCATION

Heriot-Watt University — BSc Computer Science (Artificial Intelligence)

First Class Honours, 3x Deputy Principal's Award (2022/23 – 2024/25)

BlueDot Impact — Technical AI Safety Course (2026) — [Credentials](#)

SKILLS

Languages: Python, Java, C#, TypeScript, C

Research Areas: Evaluation and reliability, generalisation analysis, retrieval systems, agentic systems

Methods and Frameworks: Transformers, Vision-Language Models, PEFT (LoRA), RAG, LangGraph

Tools & Infrastructure: PyTorch, Milvus, Docker, Linux, CUDA, Git, Weights & Biases, RunPod